Muneeb Ur Rehman

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Professional Summary

Physics graduate with hands-on expertise in Quantum Computing, Quantum Cryptography, and Machine Learning. Proven track record architecting quantum cryptanalysis protocols using Qiskit and Pennylane, simulating QKD systems, and deploying deep learning pipelines with 98% classification accuracy. Seeking to contribute to frontier research in Post-Quantum Cryptography, Quantum Communication Networks, and Quantum Machine Learning.

Technical Skills

- **Quantum Computing:** Quantum Cryptography, QKD Protocols (BB84), Post-Quantum Cryptography, Quantum Cryptanalysis, Grover's Algorithm, Variational Quantum Algorithms (VQA), QAOA, VQE, Quantum Oracles, Qiskit, Pennylane, Quantum Machine Learning (QML), Quantum Networking.
- **Machine Learning & Deep Learning:** CNN, SVM, Random Forest, Extra Trees, Ensemble Methods, Stacking Regressors, Feature Engineering, Recursive Feature Elimination (RFE), Bayesian Optimization, PyTorch, Scikit-learn, Transfer Learning (VGG16, Wav2Vec).
- **Programming Languages:** Python, FORTRAN, MATLAB.
- **Frameworks & Libraries:** Qiskit Machine Learning, Pennylane, PyTorch, Scikit-learn, NumPy, Pandas, Matplotlib, TorchConnector.
- **Software & Tools:** Linux, Wolfram Mathematica, Git, GitHub, Jupyter Notebook, Google Colab, MS Excel.

Professional Experience

Development Fellow – Physics, Quantum Communications Lab — September 2025 – Present

- Architected and simulated advanced quantum cryptography protocols, including structural quantum implementations of SPN and Feistel ciphers using Qiskit and Pennylane.
- Executed quantum cryptanalysis utilizing Grover's Search, Differential Quantum Attacks, and Variational Quantum Algorithms, reducing classical brute-force complexity from $O(N)$ to $O(\sqrt{N})$.
- Simulated the BB84 Quantum Key Distribution (QKD) protocol in Qiskit to evaluate security parameters against sophisticated intercept-resend and photon-number-splitting eavesdropping attacks.
- Validated quantum circuit fidelity across 50+ simulation runs on IBM Qiskit Aer simulator, achieving 99.2% average gate fidelity.

Machine Learning Intern, EvoSoft (Remote) — May 2024 – August 2024

- Developed an end-to-end deep learning pipeline for automated Breast Cancer Detection, processing multi-modal clinical datasets comprising 10,000+ MRI scans and X-ray images.
- Engineered and optimized Convolutional Neural Networks (CNN) and Support Vector Machines (SVM), attaining a verified binary classification accuracy of 98% and an F1-score of 0.97.
- Implemented data augmentation and cross-validation strategies, reducing model overfitting by 35% and improving generalization on unseen test data.

Summer Intern, NILOP – Islamabad — June 2024 – August 2024

- Automated chemical synthesis protocols and fabrication workflows for Magnesium-Amino Nanoparticles (Mg-Amino NPs), reducing synthesis time by 25% through standardized procedural documentation.
- Synthesized Silver Nanoparticles (Ag-NPs) optimized for photocatalytic water treatment under UV radiation, achieving 85% degradation efficiency on synthetic dye pollutants within 120 minutes.
- Conducted structural characterization via SEM and XRD analysis, documenting particle size distributions and crystallinity indices for 15+ synthesized batches.

Projects

Brute Force Attack on SPN Cipher using Grover's Search Algorithm | *Qiskit, Python, Jupyter*

- Implemented quantum simulations of Substitution-Permutation Network (SPN) ciphers, including S-AES and Mini-AES, utilizing Qiskit within Jupyter Notebook environments.
- Constructed custom quantum oracles, state-marking mechanisms, and diffuser circuits to execute quantum cryptanalysis with quadratic speedup over classical methods.
- **Results:** Successfully recovered secret keys (`c73d` & `c45f`) for a known plaintext input (`6f6d`) with 100% success rate across 20+ test iterations.

Variational Quantum Attack on 8-Bit SPN Toy Ciphers | *Qiskit, Python, Nelder-Mead*

- Designed a problem-specific, hardware-efficient ansatz in Qiskit to map secret key-space distributions for 8-bit toy ciphers.
- Optimized variational parameters using the gradient-free Nelder-Mead optimization algorithm alongside classical evaluations of S-AES and Mini-AES encryption functions.
- **Results:** Successfully isolated and recovered target cryptographic keys (`df64`) for known plaintexts (`544a`), converging within 150 optimization iterations.

Hybrid Quantum-Classical CNN Framework for Multi-Disease Detection (Qubitects) | *Qiskit ML, PyTorch, VGG16, Wav2Vec*

- Co-developed a hybrid quantum-classical prototype pitched at Pakistan's 1st Quantum Computing Hackathon, targeting lung, heart, skin, and breast diseases.
- Utilized VGG16 and Wav2Vec feature extractors for classical processing, transferring structural weights into quantum encoding modules featuring ZZFeatureMap and Amplitude Encoding.
- Compiled and trained the hybrid architecture via the PyTorch `TorchConnector` module within Qiskit Machine Learning; deployed on a local server for testing, achieving a 75% classification accuracy across 4 disease categories.

Ensemble Machine Learning for Glass-Forming Ability of Bulk Metallic Glasses (FYP) | *Python, Scikit-learn, Bayesian Optimization*

- Engineered an ensemble machine learning framework in Python to predict the glass-forming ability of BMG alloys based on simulated and literature-sourced data comprising 5,000+ alloy compositions.
- Executed Recursive Feature Elimination (RFE) for dimension reduction from 12 to 6 key features, Bayesian optimization for hyperparameter tuning, and a Stacking Regressor meta-model architecture.
- **Results:** Achieved robust predictive performance with R-squared = 0.93, RMSE = 1.42, and MAE = 0.77, outperforming baseline Random Forest by 12%.

Machine Learning Framework for Predictive Material Yield Strength | *Python, Pandas, Scikit-learn, Google Colab*

- Curated, preprocessed, and engineered features from open-source material science datasets using Python, Pandas, and MS Excel, processing 3,000+ material records.
- Evaluated multiple regression architectures in Google Colab, determining that Random Forest and Extra Trees Regressors yielded the peak accuracy profile with R-squared = 0.97 and RMSE = 2.1.

Deep Learning-Based Medical Breast Cancer Detection | *Python, Keras, TensorFlow, SVM*

- Formulated a computer vision pipeline on Kaggle medical image datasets using Convolutional Neural Networks (CNN) and Support Vector Machine (SVM) classifiers, processing 9,000+ histopathology images.
- Achieved a 98% binary classification accuracy score for early-stage malignancy recognition, with sensitivity of 96% and specificity of 99%.

Education

Pakistan Institute of Engineering and Applied Sciences, Bachelor of Science in Physics — September 2021 – July 2025

- **Relevant Coursework:** Computational Physics, Quantum Mechanics, Fundamentals of Machine Learning, Electromagnetic Theory, Linear Algebra, Multivariable Calculus, Solid State Physics, Statistical Mechanics.

Punjab Daanish Boys School (DG Khan), Higher Secondary School Certificate (FSc Pre-Medical) — August 2019 – July 2021

Punjab Daanish Boys School (DG Khan), Secondary School Certificate (Matriculation in Science) — August 2017 – July 2019

Certifications

International Training Workshop on Quantum Computing & Quantum Information – *COMSTECH, OIC* — May 2026
QKD and BB84 Protocols: Theory to Simulation – *QRISE, NUST* — February 2026
Workshop on Quantum Computing – *The Qubit Pakistan* — November 2025
QNickel Quantum Computing Certification Course (OQI Hackathon) – *QWorld* — August 2025
QBronze Quantum Computing Certification Course (OQI Hackathon) – *QWorld* — August 2025
The Science of Success: What Researchers Know That You Know – *University of Michigan* — March 2025
Basics of Machine Learning – *Great Learning* — June 2024
Introduction to Exploratory Data Analysis Using Excel – *Great Learning* — September 2023
Excel Basics for Data Analysis – *IBM* — April 2023
Crash Course on Python – *Google* — March 2023

Awards & Honors

National Finalist, 1st Quantum Computing Hackathon – Pakistan — February 2026

- Selected as a national finalist in Pakistan's premier quantum computing competition, organized jointly by the National Center for Physics (NCP) and the Open Quantum Institute (OQI) of CERN.
- Developed and pitched the *Qubitects* hybrid prototype designed for remote multi-disease evaluation, explicitly targeting UN Sustainable Development Goals (SDGs) for healthcare localized to regional challenges.

Letter of Appreciation in Academics – *PDBS, DG Khan* — September 2019

- Awarded by the institution administration in recognition of attaining distinction standing during the secondary matriculation examinations.

Leadership & Activities

Co-Organizer, 1st International Workshop on Recent Advances in Quantum Science – *CMS, PIEAS* — February 2025

- Co-organized and contributed to a comprehensive 5-day technical seminar program evaluating structural advancements in fundamental quantum applications and proton neural network infrastructures.
- Coordinated speaker logistics and managed attendance of 200+ participants from 10+ national institutions.

Management Team Member – *PIEAS Physics Society* — August 2022 – Present

- Coordinated logistical orchestration for academic outreach events targeted at introducing undergraduate demographics to topics across classical mechanics and quantum field principles.
- Managed event budgets and volunteer teams of 15+ students for 8+ semesterly physics seminars and workshops.